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Full-waveform simulation of DAS records, response and cable-ground coupling

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SUMMARY

Over the past several years, the use of optical fibre cables as ground motion sensors has become a central topic for seismologists, with successful applications of distributed acoustic sensing (DAS) in various key fields such as seismic monitoring, structural imaging and source characterization. DAS response is a combination of both instrument response and cable-ground coupling, with the latter having a strong, spatially variable, but yet largely unquantified effect. This limits the application of a large number of staple seismological techniques (e.g. earthquake magnitude estimation, waveform tomography) that can require accurate knowledge of a signal's amplitude and frequency content. Here we present a method for accurately simulating a DAS cable and its ground coupling. The scheme is based on molecular dynamic-like particle-based numerical modelling, allowing the investigation of the effect of varying DAS-ground coupling scenarios. We start by computing the full strain field directly, for each pair of neighbouring particles in the model. We then define a virtual DAS cable, embedded within the model and formed by a single string of interconnected particles. This allows us to control all aspects of the cable-ground coupling and their properties at an effective granular level through changing the bond stiffness and bond types (e.g. non-linearity) for both the cable and the surrounding medium. Arbitrary cable geometries and heterogeneous materials can be accommodated at the desired scale of investigation. We observe that at the meter scale, the cable-ground coupling and local site effects can substantially alter the recorded signal. We find that the stiffness of the thin layer of material to which the cable is coupled has the strongest effects, selectively amplifying portions of the wave train and contributing to substantial phase delays. These differences show that cable coupling and local site effects should be considered both when designing a DAS deployment and analysing its data when either true or along-cable relative amplitudes and/or frequencies are considered. The codes developed herein for calculating full waveform DAS responses and coupling are made publicly available.

Key words: Computational seismology; Distributed acoustic sensing; Seismic instruments; Site effects; Wave propagation.

1 INTRODUCTION

The first reported applications of optical fibre cables as ground motion sensors in exploration geophysics started under the name of distributed acoustic sensing (DAS) in the early 2010s (e.g. Mestayer *et al.* 2011; Molenaar *et al.* 2012; Daley *et al.* 2013), and were followed by a large number of successful, pioneering applications in seismology since 2017 (e.g. Lindsey *et al.* 2017, 2019; Jousset *et al.* 2018; Marra *et al.* 2018; Wang *et al.* 2018). DAS can provide spatial resolution of less than a metre, and has a sensitivity bandwidth ranging from quasi-static deformation to MHz (Hartog 2017). This, together with the relative ease of deployment and the large networks of telecommunication fibres already potentially available for interrogation (Marra *et al.* 2018), gives DAS the potential to fill the gap in scenarios where traditional ground motion sensors are either lacking sensitivity (very low, very high frequencies), or are difficult to deploy—for example volcanoes or ocean floor (Jousset *et al.* 2022; Marra *et al.* 2022).

One of the main obstacles to the application of DAS, however, is that its instrument response is still largely unknown (Lindsey *et al.* 2020). This prevents the application of seismological techniques that require accurate absolute amplitudes or a detailed understanding of the spectral characteristics of the recorded signals. These range from staple seismological applications such as earthquake magnitude estimation and full-waveform inversion to cutting-edge processing on time-lapse studies that target extremely small waveform variations.

How seismic sensors respond to ground motion is a combination of multiple factors that include instrument design, instrument-ground coupling and local ground site effects. In DAS cables, instrument design is both determined by the DAS interrogator characteristics (Gabai & Eyal 2016), the cable's finite size and strong directional sensitivity (Martin 2018) and the mechanical properties of the fibre itself. For sub-g accelerations, seismometers are ground coupled by their own weight, while strain on a DAS cable has a dependency on frictional contact with the ground. As cables are often buried in trenches, unconsolidated trench backfill may thus play a significant local site effect role. In addition to this, both coupling and local effects may vary along the cable length.

In the past, the strong effect that different types of fibre installation have on the DAS records was observed in vertical seismic profiling (VSP), but its characteristics were not quantified (Pevzner *et al.* 2021). Recently, independent seismological studies (Lindsey *et al.* 2020; Paitz *et al.* 2021; Harmon *et al.* 2022) approached the problem empirically, systematically estimating the response of a DAS cable in different experiments by comparing it to co-located seismometers with known response. These studies observed large changes in the instrument response for fibres mostly decoupled from the ground—for example fibre loosely suspended on grass (Harmon *et al.* 2022)—but only relatively small changes for fibres coupled to the ground using different methods such as sand bags, tape and wooden blocks (Paitz *et al.* 2021). These empirical responses however, are estimated after converting the strain rate signal to displacement or velocity, and assumptions in the processes of conversion and comparison may well contribute to response variations (Paitz *et al.* 2021). In addition, DAS is still an emerging technique and deployments are relatively few, so that we lack data for a comprehensive empirical approach. It is also important to note that some of the most beneficial applications of DAS are for scenarios where reference seismometers cannot be readily deployed, for example Ocean floors (Marra *et al.* 2022) or active volcanoes (Jousset *et al.* 2022).

In this study, we aim to systematically analyse the influence of cable-ground coupling and local ground heterogeneities (i.e. local site effects) on the strain field recorded using DAS. In order to help generalize the findings of empirical studies and to provide a means of quantifying the contributions to DAS response, we approach this in a synthetic framework, where we can introduce changes to the experiment one at a time and single out their individual effects. By quantifying the effects of different aspects of the cable-ground interaction, we aim to provide key information on how to interpret some common observations on DAS data as well as to highlight best practices for future DAS deployments.

2 MODELLING DAS

One way to fully understand the influence of site effects and cable-ground coupling on DAS records is to model their effects on the propagating wavefield. In particular, we aim to reproduce lateral variations in the medium (i.e. local site effects), cable properties (e.g. cable material and trenching) and cable-ground coupling (i.e. cable-ground bond). Additionally, DAS cables are mostly sensitive to components of ground motion that are parallel to the cable's axis (Kuvshinov 2016), so it is essential to use a modelling tool that can handle the often complex geometries of realistic DAS deployments.

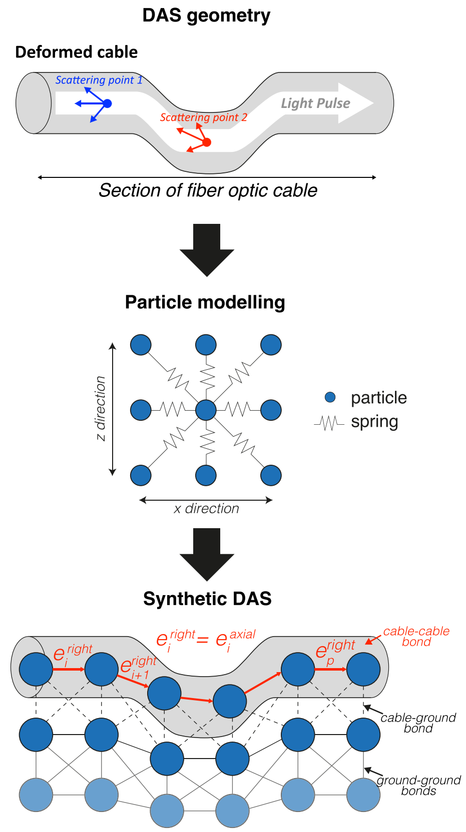

Figure 1. Schematic of particle modelling of a DAS cable. On the top, a cartoon of a deformed section of a DAS cable. In the middle we show the geometry of the square elastic lattice used by ELM2D-DAS. On the bottom, we show how ELM2D-DAS can be used to model the deformed cable and the surrounding media.

To accurately model the complex material properties necessary to simulate differently coupled cables, we use a numerical waveform modelling scheme based on discrete particle schemes (DPS; Monette & Anderson 1994; Toomey & Bean 2000; Toomey *et al.* 2002; O'Brien & Bean 2004). These models enable us to easily control the material properties of each model particle and its bonds to its neighbours on an effective granular level (Fig. 1).

2.1 Particle modelling

Particle models have been used to simulate the extreme complexity of geomaterials, modelling them as a tight lattice of particles that interact based on Hooke's law—the discrete particle scheme (DPS; e.g. Toomey & Bean 2000). They were initially developed to simulate crystal lattices at an atomic scale and are widely used in geomechanics, where particles are sized to represent sediment grains or larger 'intact' units of rock. In seismology, the required effective granular scale is directly related to the target wavelengths (Toomey & Bean 2000; typically several metres), and DPS and a sister scheme called the elastic lattice method (ELM) have been successfully used to model waveform propagation (Toomey & Bean 2000; O'Brien & Bean 2004) and simulate complex material properties such as fracturing (Toomey *et al.* 2002), viscoelasticity (O'Brien 2008) and non-linearity (O'Brien 2021).

In this study, we develop ELM2D-DAS, a numerical particle modeller that allows the simulation of a DAS cable and its strain

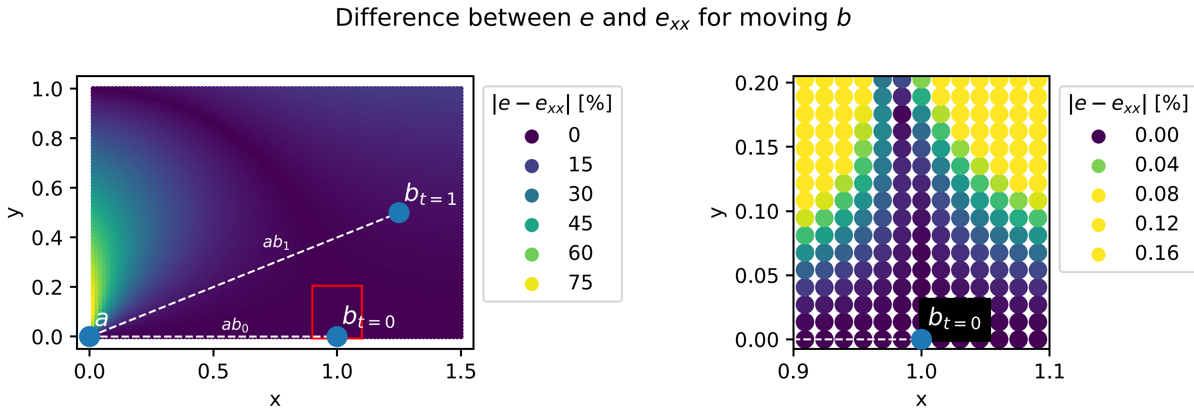

Figure 2. Difference between bond strain e and horizontal strain e_{xx} between two particles with changing position. Left-hand panel shows the position of particles 'a' and 'b'. The initial position of 'b' is denoted as '$b_{t=0}$', while its position after being displaced is both denoted as '$b_{t=1}$' for a single case, and as a dense scatter plot for different displacements. The colour of the scatter represents the difference between e and e_{xx}. Right-hand panel shows a zoom on the area boxed in red.

output in 2-D. The code is largely based on the ELM scheme, following (O'Brien & Bean 2004; O'Brien 2021), which we expand to compute strain, include a virtual DAS cable, and wrap within a Python framework to boost flexibility in parallel computing environments.

In ELM, the model is composed of a square lattice of particles, linked by a combination of linear and non-linear springs to simulate both linear and non-linear elastic and viscoelastic media (O'Brien 2021). The spring's stiffness is computed from the desired properties of the material (e.g. its elastic wave velocities) and their elastic and bond-bending constants define the response of the medium to both dynamic and quasi-static stress perturbations (O'Brien & Bean 2004).

In contrast to most available numerical waveform propagation methods, ELM solves the equations of motion for each particle rather than the wave equation: given an input lattice made of particles connected by springs, and a force source, the code computes the particle position at the next time step deriving the displacement from Hooke's law (O'Brien & Bean 2004). The scheme leads to full wavefield propagation. Because we can express the Lamé parameters in terms of spring and bond-bending constants (O'Brien & Bean 2004), we can model heterogeneous media by locally changing the properties of the model's particles and attached springs. Additionally, we can easily simulate changes in the bond stiffness (weak coupling, fractures; Toomey *et al.* 2002) and deviations from linear elastic behaviour (O'Brien 2008, 2021) by altering the coefficients of the linear and non-linear springs, respectively.

2.2 Modelling strain in lattices

In order to model the quantities that are inherently sensed by DAS systems, we augment ELM by implementing the computation of strain. Computing strain in lattices can be challenging (O'Sullivan *et al.* 2014), as strain is often expressed as the deformation of a cell rather than in terms of discrete particle positions (Haxha & Migliorato 2010). Monette & Anderson (1994) provide a relationship to compute the strain components from displacement in 2-D. In addition to this, we have seen that particle models are based on particle–particle interactions, so that instead of measuring the change in length of an element, we can also measure the bond strain, that is the change in length of the bond between two particles. Bond strain is particularly suited to our study, where we ultimately want to simulate the axial strain measured by an arbitrarily oriented DAS cable rather than the strain field along the x- and y-coordinates.

In a square lattice geometry such as the one used by ELM2D-DAS, for a central particle i and its eight neighbours $n,$ we can compute the bond strain as:

$$e_i^n = \frac{L_n^t - L_n^{t_0}}{L_n^{t_o}}; \tag{1}$$

with n the neighbour index and L_n^t, $L_n^{t_0}$ the distance between particles at the time step t and at the initial unperturbed distance, respectively. Although the bond strain in a square lattice has eight components, it can be fully represented by only four of them, as they are symmetrical around each bond: $e_i^{\text{right}} = e_{i+1}^{\text{left}}$ for two neighbouring particles at $x = i,\ i+1$. This holds true both in homogeneous and heterogeneous media, as the strain on each bond results from the displacement of both particles forming it, which is computed taking account of each particle's spring (and thus elastic) constants.

It is important to note that bond strain is measured in the direction of the bond rather than along the x-, y-axes, that is it measures the modulus of the strain e rather than e_{xx} or e_{yy}. In the case of small displacements, however, $e - e_{xx}$, for instance, is very small, with different combinations of vertical and horizontal displacement up to 1 per cent of the particle distance yielding a negligible difference between $e - e_{xx}$ (i.e. <0.1 per cent, see Fig. 2). Since in our models the particle distance is typically 1 m, displacements as large as ~1 cm would yield negligible difference between e and e_{xx}. The same applies to e_{yy}.

In Fig. 3, we simulate a simple lattice unit (one central particle and its neighbours) and change the position of the central particle i and its neighbours arbitrarily to measure the resulting bond strain. The results show how the bond strain field for all neighbouring particles changes when only the centre particle is displaced.

We implement the same scheme in ELM2D-DAS, where, at each time step, we compute the strain between each particle and its eight neighbours following eq. (1). This provides us with the complete bond strain field at each time step, alongside displacement and velocity. By computing a simple model consisting of a 10 km × 10 km homogeneous half-space (V_P 3000 m s^{-1}, V_S 2000 m s^{-1}, density 2000 kg m^{-3}) with 10 m particle distance, a free surface

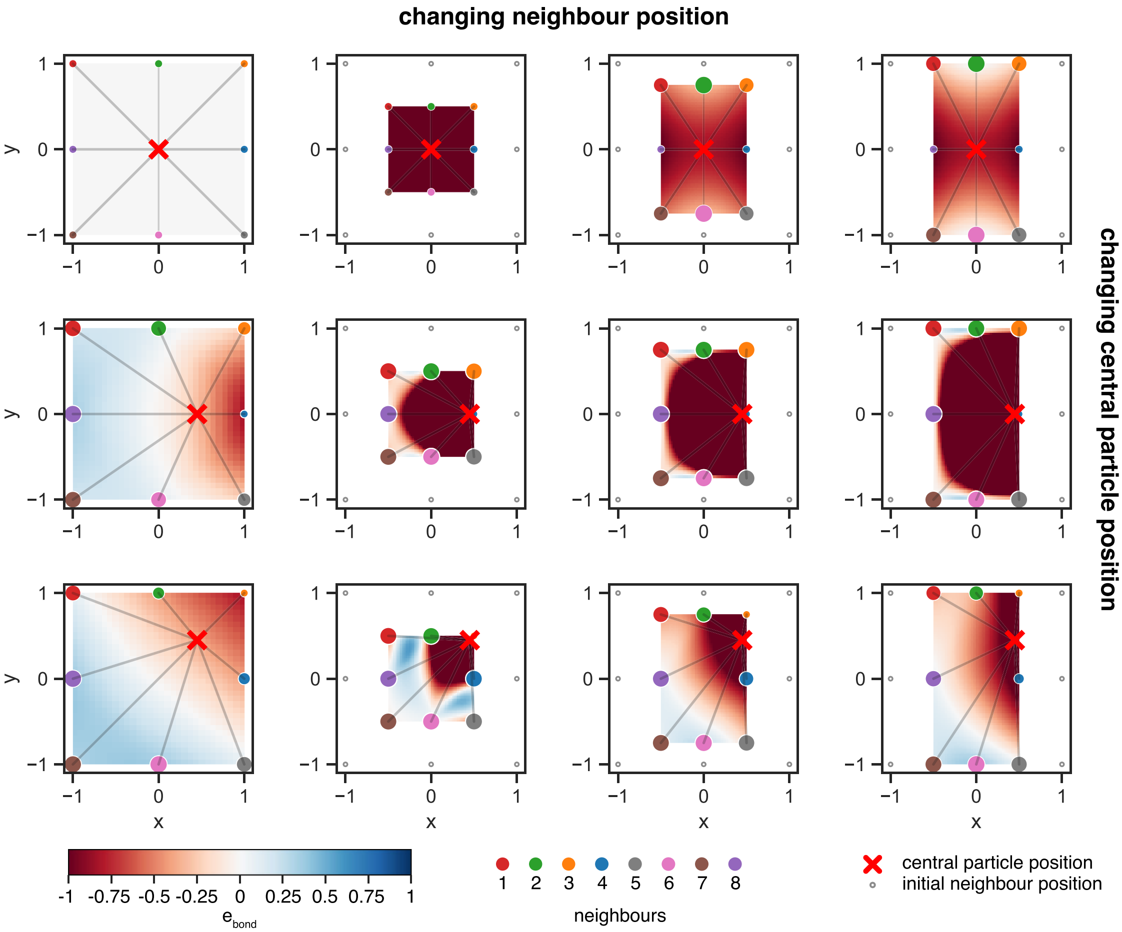

Figure 3. Bond strain field in a square lattice geometry. The bond strain is shown as a continuous field interpolated from the strain computed between the central particle and each of the eight neighbours. We show the changes in the bond strain field for both different central (change in between rows) and neighbouring particle positions (change in between columns). Central particle position is a red cross and neighbouring particle positions are circles, coloured by neighbour and with size depending on the magnitude of their strain.

and a 5 Hz explosive source (Morlet wavelet source–time function), we can observe the different components of the propagating strain (Fig. 4).

2.3 Modelling a DAS cable

After implementing a measure of the strain field, we simulate a DAS cable that is embedded within the medium. We implement this by defining the cable as composed of a single string of interconnected particles, allowing us to control its properties.

DAS measures the strain (or strain rate depending on the interrogator) along the cable's axis (Hartog 2017); in ELM2D-DAS, this is the strain e_i^n between the ith particle and the bond to its nth neighbour that are part of the simulated cable, so that $e_i^{\mathrm{axial}} = e_i^n$ (Fig. 1). In the case of a horizontal DAS cable, this will be e_i^{right} or e_i^{left}, while for a vertical cable, it will be e_i^{above} or e_i^{below}. The effect of measuring over gauges can be easily implemented by averaging the bond strain for all particles within a given gauge length; the axial strain over a single gauge can thus be measured as

$$e_{\mathrm{gauge}} = \sum_{i=0}^{p} e_i^{\mathrm{axial}}/p; \qquad (2)$$

with p the total number of particles within each gauge and e_i^{axial} the strain between the ith particle and its neighbour on the simulated DAS cable.

This design allows us to easily simulate complex cable geometries, where the cable can change direction multiple times even within a single gauge, sensing different components of the background strain field. In ELM2D-DAS, we measure the axial strain e^{axial} for each particle along the cable, so that our capability to account for changes in cable direction is limited only by the discretization of the model.

In addition to simulating the complex cable geometries of realistic DAS deployments, this scheme also allows us to take into account the dynamic changes in the cable direction that happen when the cable is deformed by the incoming wavefield (as shown by the exaggerated particle displacement in Fig. 5). Although these are likely negligible for most cases (Fig. 2), they could be relevant in strong ground motion scenarios, where large displacements could dynamically alter the cable geometry and thus its directional sensitivity ($e^{\mathrm{axial}} \neq e_{xx}$).

In Fig. 5, we show the DAS record produced by a synthetic cable in the same simulation shown in Fig. 4. The simulated DAS cable is positioned at the free surface but is also inset

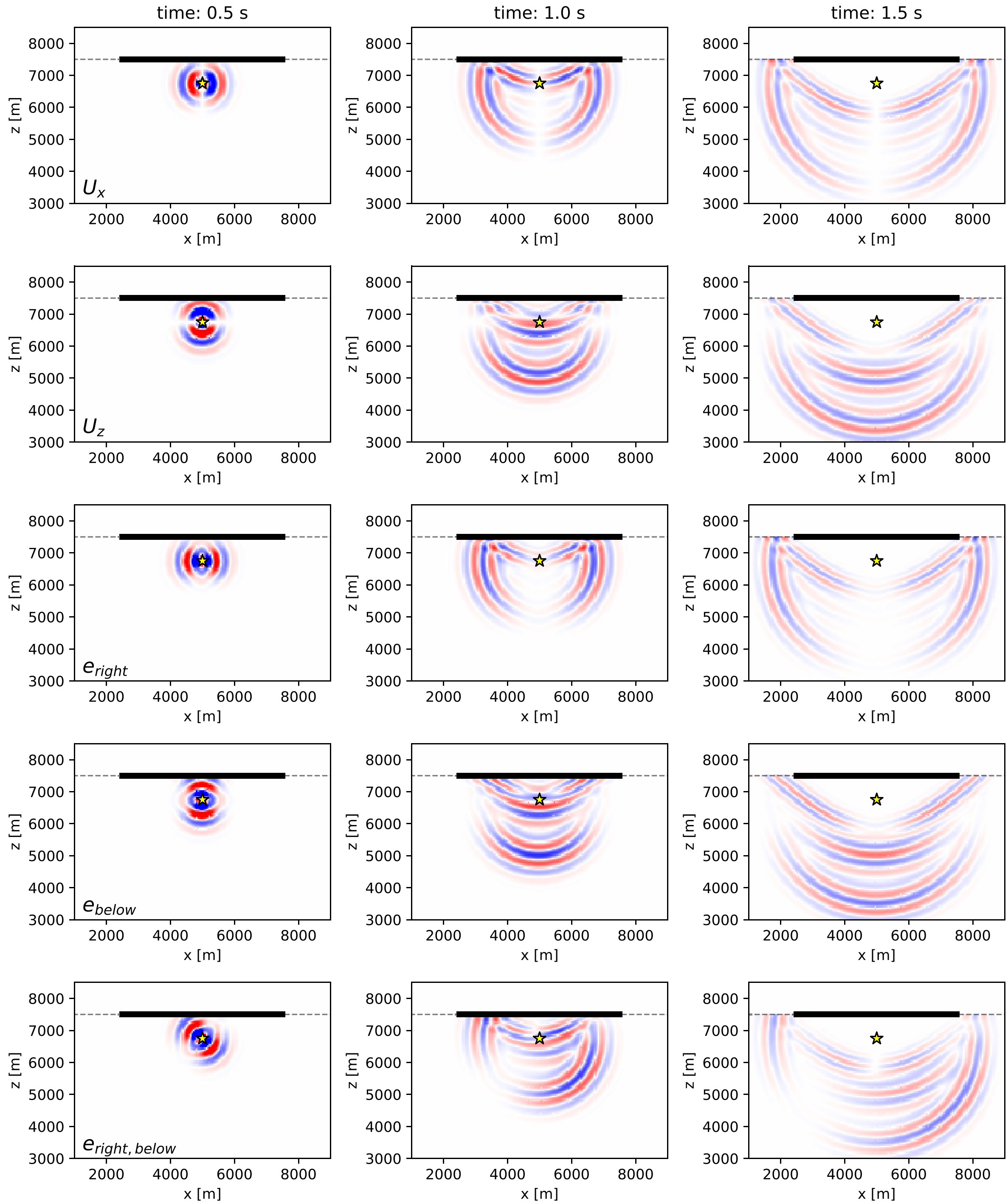

Figure 4. Full wavefield simulation of displacement and strain for an explosive source. Rows show the horizontal and vertical components of displacement and the bond strain for the right, bottom and bottom-right bonds. Columns show snapshots of the wavefield at different times. All amplitudes are normalized to the maximum value at all times for each simulation (excluding the near source area). In all panels, the virtual DAS is shown as a bold line, the free surface as a dashed line and the source location as a yellow star.

into the ground, with the top of the cable exposed at the free surface. The cable has a length of 5 km and a gauge length of 100 m, without any channel overlap. In this simplified example we use a longer-than-average gauge length to include the effect of gauge-averaging with the relatively coarse particle distance (10 m). Because the cable is horizontal, $e_i^{\mathrm{axial}} = e_i^{\mathrm{right}}$ and we show u_x (averaged along each gauge) for comparison.

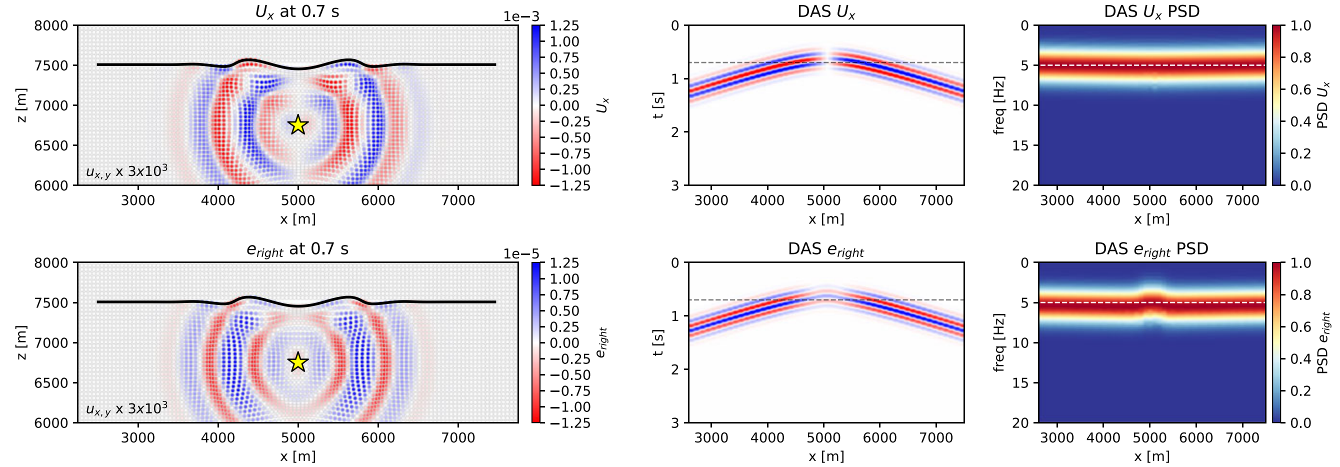

Figure 5. Waveform propagation of strain and displacement fields and their synthetic DAS records. The displacement of the particles in the snapshots is exaggerated by 3e^3 for visualization purposes. The simulation is run with an explosive source 5 Hz Ricker wavelet in a 10 km × 10 km medium with a free surface and a synthetic DAS beneath it. We show the horizontal displacement (top panel) and right-bond strain (bottom panel) both as snapshot of waveform propagation at 0.7 s (left-hand panel) and as recorded by a synthetic DAS with gauge length of 100 m (right-hand panel). Synthetic DAS cable location is shown as a black line and the source as a yellow star. Synthetic DAS records are shown as time cross-sections for both horizontal displacement and axial strain (i.e. right-bond strain). A dashed line marks the time at which the wavefield snapshots on the left are taken. Snapshot for the full waveform simulations can be found in Fig. 4.

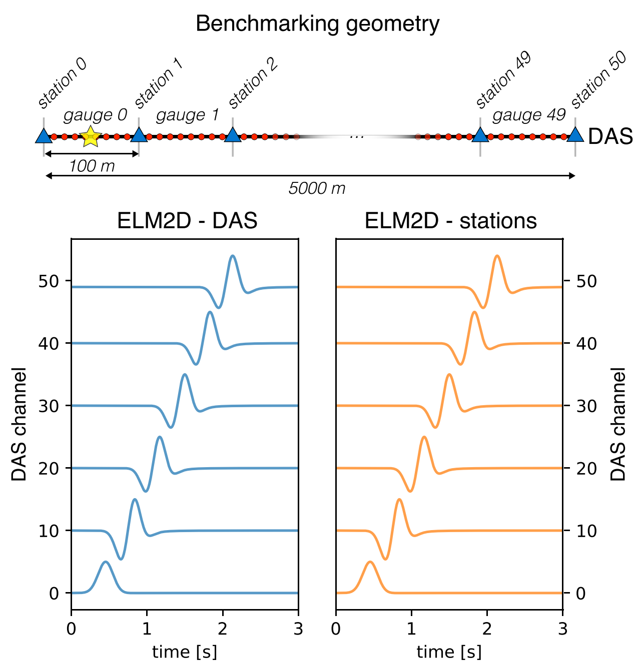

Figure 6. Benchmarking of axial strain records from ELM2D-DAS. The experiment geometry (identical for all simulations) is shown on the top: thin vertical lines are the limits of the DAS gauges; the source is the yellow star and stations are shown as blue triangles. Axial strain (ELM2D-DAS) and horizontal strain (ELM2D-stations) records are shown for gauges 0, 10, 20, 30, 40 and 49. Horizontal strain was computed differentiating the displacement at stations at each gauge's end (Kennett 2022).

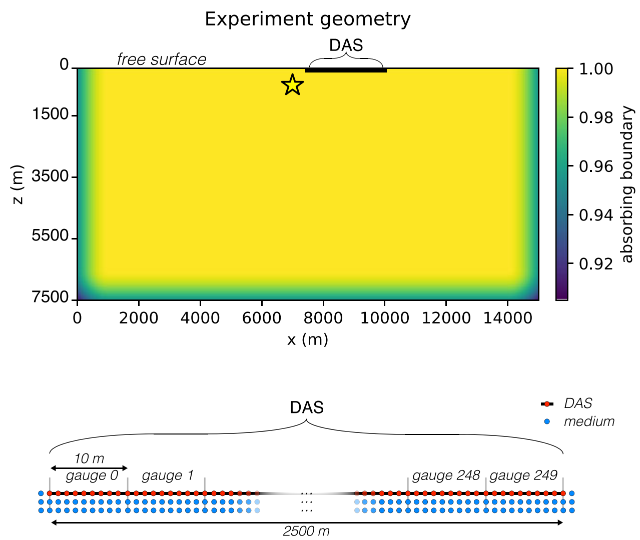

Figure 7. Geometry of the experiment used throughout Sections 3.1–3.3. Top panel shows the model geometry, with the colour representing absorbing boundary conditions. Bottom panel shows a detailed schematic of the DAS cable geometry, with gauge lengths (thin grey vertical lines) and particles forming the simulated DAS cable and the medium directly surrounding it (red and blue circles, respectively. The source location is plotted as a yellow star and the simulated DAS cable as a solid black line.

2.4 Benchmarking

In this section, we benchmark the DAS strain records from ELM against other, comparable measures. To do this, we define a very simple setup with a horizontal 5-km-long virtual DAS starting at the centre of a 20 km × 20 km homogeneous medium, with 100 m gauge length and an explosive Gaussian source at the centre of the first gauge (Fig. 6).

2.4.1 *Station differentiation*

We compare our synthetic DAS output to the displacement measured by ELM using the station differentiation technique, in which the average strain over a gauge is computed by differentiating the displacement measured at the gauge's ends (Mateeva *et al.* 2014;

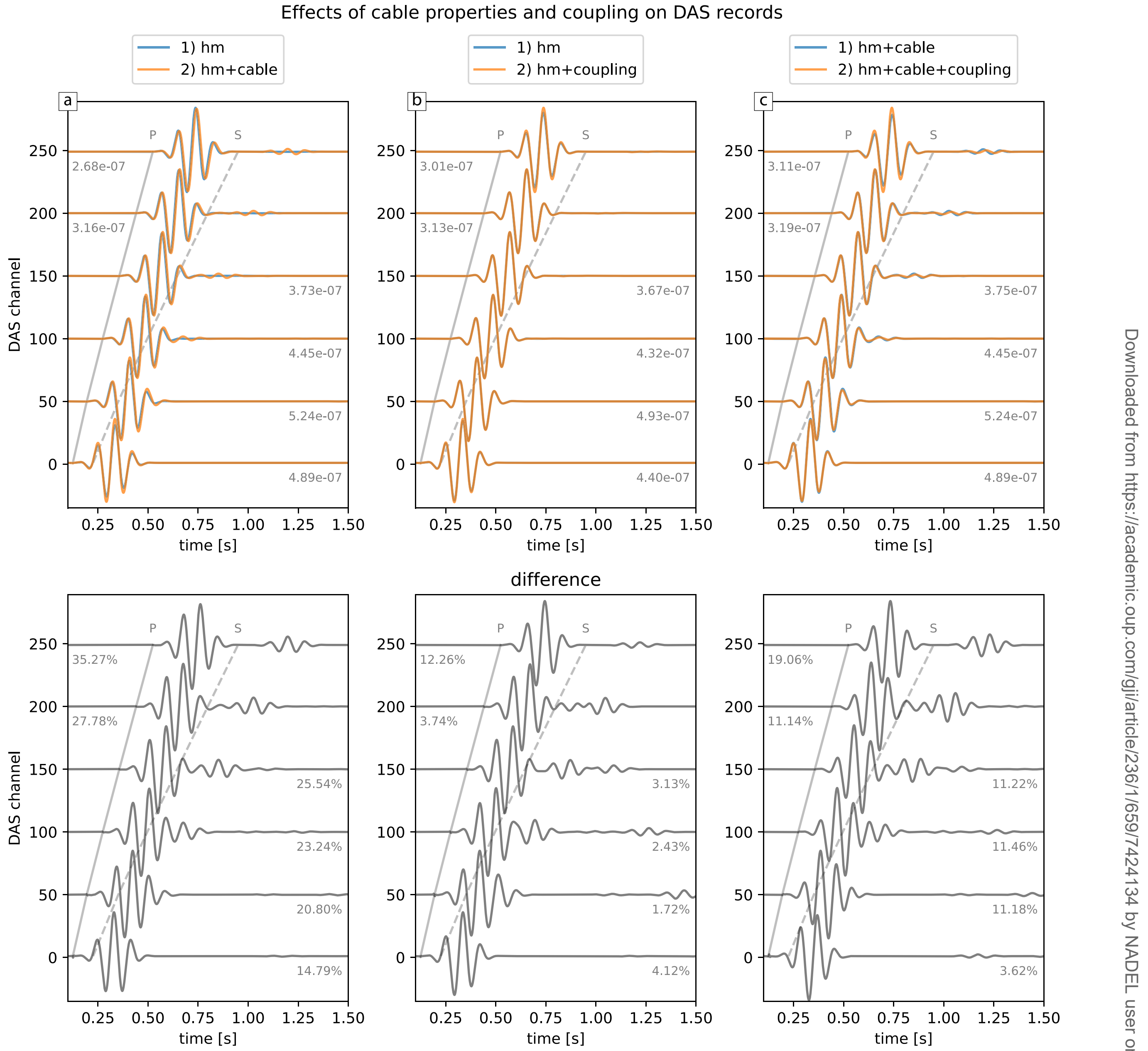

Figure 8. Effect of cable properties and coupling on DAS records. The geometry of the experiment is described in Fig. 7. Each column represents a simulation: top, DAS channel axial strain records (orange), compared to the reference case (blue) (legend for both records plotted on top of each panel); bottom, difference between the two simulations. In all panels, hm indicates homogenous half-space. All channel records are normalized to the maximum axial strain amplitude, which is reported next to each waveform. The amplitudes of the differences are shown as percentage of the axial strain maximum.

Bakku 2015; Kennett 2022). For a horizontal cable where $u_{\mathrm{axial}} = u_x$:

$$e_{\mathrm{gauge}} = \frac{\Delta u^x_{\mathrm{station}}}{l_g} = \frac{u^x_{i+1} - u^x_i}{l_g}; \tag{3}$$

with l_g the gauge length. By including in ELM2D-DAS virtual seismic stations measuring u_x at the end of each gauge, we see (Fig. 6) that the strain measured directly by ELM2D-DAS is identical to the strain obtained from the differentiated displacement. Since the displacement from ELM has already been benchmarked against finite difference methods (O'Brien & Bean 2004) and Analytical solutions (O'Brien 2008), this comparison validates our new implementation of strain in ELM2D-DAS.

2.4.2 *Advantages of ELM2D-DAS*

It is worth noting that the method we use to validate our results has limitations when it comes to simulating DAS strain records: the station differentiation technique can only be used at the local scale (Kennett 2022) and becomes difficult to apply with complex cable geometries. ELM2D-DAS is capable of computing the axial strain in a cable directly, while also allowing the easy implementation

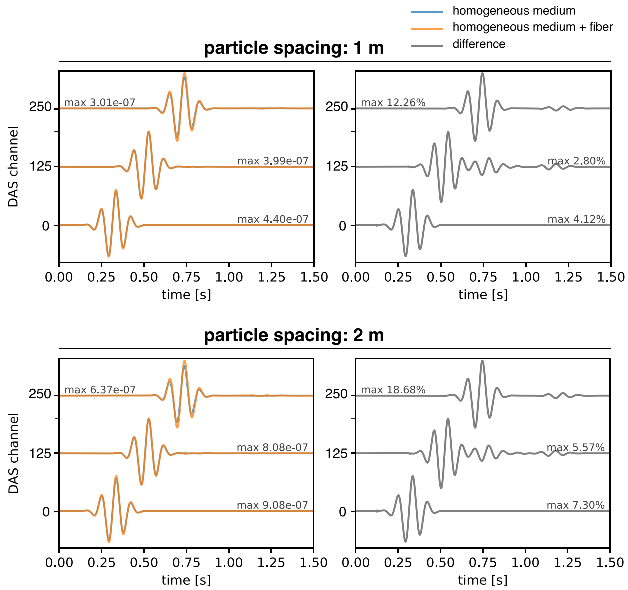

Figure 9. Effect of changing the effective cable width on the recorded wavefield. Top panels show the simulation from Fig. 8(a). Bottom panels show the same setup with a coarser particle distance of 2 m. Left-hand panels show the simulations of a simple homogeneous medium (blue) and a homogeneous medium plus a DAS cable with its own material properties (orange). Right-hand panels show the difference between the two simulations.

of complex material properties and fractures at effective granular scales.

3 MODELLING CHANGES IN DAS RESPONSE

In the previous sections, we demonstrated how we can use particle schemes to reliably compute strain in a medium and simulate the record of a DAS cable embedded within it. Now that we can accurately simulate the physics of DAS records, we can use this solver to quantitatively analyse which cable-ground interaction scenarios may affect the response of a DAS cable, and to what extent.

A seismogram can be typically described by three terms that are convolved together: a source term, the instrument response and a Green's function describing the medium through which the wavefield travels. In the case of a traditional seismometer, the instrument response is largely thought of as controlled by the instruments' design, while its ground coupling and the properties of the medium in its immediate surroundings (local site effects) are generally considered separately and addressed qualitatively in terms of how 'noisy' the site is. For the case of sub-g accelerations, seismometers are usually ground-coupled through gravity. This is not true for a DAS cable, where both coupling and site effects can vary along the cable length.

In this section, we will use ELM2D-DAS to simulate how subtle changes in the cable-ground coupling and local site effects can affect the DAS record, exploiting the granular control the ELM solver gives over medium and bond properties.

In order to do this, we perform a range of simulations testing the effects of different changes in material properties and bond types on the virtual DAS record. To discern the effects clearly, all simulations share the same simple model design: a 15 km × 7.5 km homogeneous half-space with 1 m particle spacing and $V_P = 5800\ \mathrm{m\,s^{-1}}$, $V_S = 3200\ \mathrm{m\,s^{-1}}$ and $\rho = 2600\ \mathrm{kg\,m^{-3}}$. The model includes a 2.5 km horizontal DAS cable inset in the medium at the free surface, with a 10 m gauge length. The source is offset from the cable's start by 500 m both vertically and horizontally and is explosive, with a 10 Hz Morlet time function. This setup is depicted in Fig. 7. Notably, this geometry has no heterogeneity other than the ones we will introduce around the cable in the following sections. This geometry is likely representative of a wide range of real DAS applications (not teleseisms), in which the cable is at the surface and the source is nearby at depth. We will discuss the effect of different setups in later sections.

Because the lattice geometry used is regular, we need to balance the particle spacing (required to accurately capture the small heterogeneities around the DAS) and the model size. The settings described above require a simulation with over 112 million particles, which we can run over 150 cores for an average of 130 min per core, per simulation on a Linux server equipped with an Intel Xeon Gold 6148 2.40 GHz processor with 160 cores and 1.47 TB RAM to output 2 s seismograms. In the next sections, we detail the effects on the recorded waveforms of each of the changes in the cable-ground scenario.

3.1 Cable material properties and coupling

At first, we want to test the effects on DAS records of the cable's elastic properties and its coupling to the medium.

3.1.1 *Cable material properties*

Because our simulated DAS cable is embedded in the medium, we can easily assign different V_P, V_S and ρ values to the particle forming the cable. The elastic properties of optical fibres however are not readily available from manufacturers and will also vary if we consider factors such as fibre cladding, armouring and variably packed groups of fibres. It is however clear that fibres do not have the same elastic properties as solid rock. For this reason, we assign $V_P = 3000\ \mathrm{m\,s^{-1}}$, $V_S = 2000\ \mathrm{m\,s^{-1}}$ and $\rho = 2000\ \mathrm{kg\,m^{-3}}$ to the string of particles forming the DAS cable so that it roughly matches the properties of silica glass. Our simulated cable is formed by a single string of particles. As the grid spacing is 1 m, this leads to a cable of the same effective thickness.

Compared to the simulations in a homogeneous half-space, simulating a cable with its own elastic properties only has subtle, yet notable effects (Fig. 8a): the DAS channels close to the source are visibly amplified, and all channels show a very small delay. More importantly, we observe the appearance of clear surface waves in all traces, which are invisible to the naked eye in the homogeneous medium simulation due to their extremely low amplitude. Our observations indicate that the material properties of the fibre have little effect on the body wave arrivals but visibly amplify surface waves. Although the amplitude of these surface waves is relatively small due to the source type (explosion) and depth (source not at the surface), their presence indicates that the material of the cable has a clear, non-negligible effect on the recorded seismograms.

While a simulation of a thin (e.g. 0.02 m) cable is not computationally feasible with our current setup, we can measure the influence of the cable thickness by measuring how our observed effects change for a coarser grid and a thicker cable. We compare our simulation with 1 m particle spacing to an identical simulation geometry where the particle spacing is 2 m (i.e. a thicker cable, Fig. 9). Interestingly, we observe that adding a cable with its own material properties has a nearly identical effect independent of the particle

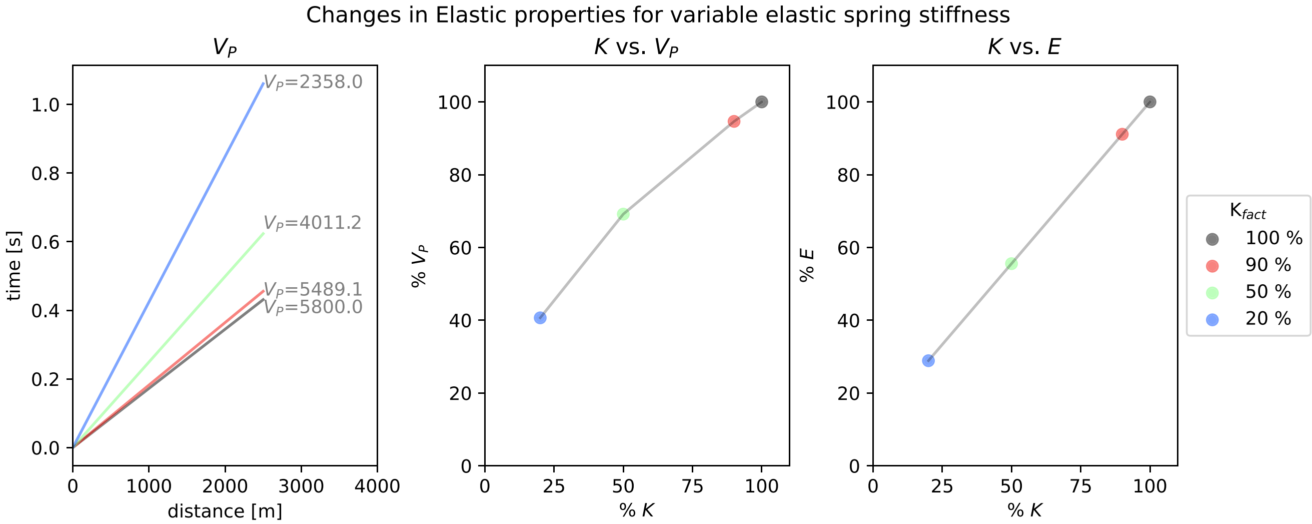

Figure 10. *P*-wave velocity and Young's modulus for a variable spring stiffness. Left-hand panel shows the changes in V_p resulting from a spring constant $\boldsymbol{K} = \boldsymbol{K}_{\mathbf{fact}} * \boldsymbol{K}_0$ for variable $\boldsymbol{K}_{\mathbf{fact}}$. Middle and right-hand panels show the relationship between $\boldsymbol{K}_{\mathbf{fact}}$ and the resulting change in V_p and Young's modulus E, respectively, expressed as percentage of the values for a strong medium. To compute the Young's modulus from the Lamé parameters, we used a Poisson's ratio of 0.25.

spacing, with the 2-m-spaced simulation showing the same exact wave shape as the 1 m one, with a very small delay of 0.0004 s at the furthest channel. Importantly, both simulations show the excitation of identical surface waves. This test indicates that the thickness of the simulated cable does not significantly affect our results for wavelengths much larger than the fibre cable width and could become significant only when investigating frequencies of hundreds of Hz.

3.1.2 Ground-cable coupling

As second test, we analyse the effect of changing the cable-ground coupling. In ELM2D-DAS, we can easily achieve this by changing the stiffness of the linear spring K at the medium-cable interface so that $K = K_{\mathrm{fact}} * K_0$, and choosing values of K_{fact} smaller than one. In our simulations, we significantly weaken the bond so that $K_{\mathrm{fact}} = 0.2$ to simulate a 'worst-case' scenario, which is a very weak bond approaching no coupling (Toomey *et al.* 2002). In Figs 8(b) and (c) we show the effects of weak cable-ground coupling for both a cable sharing the same elastic properties as the medium and a cable with its own elastic properties as described in Section 3.1.1.

Cable weakly coupled to a homogeneous medium and sharing the same material properties: In the case of a cable sharing the homogeneous medium's elastic properties (Fig. 8b), weakening the cable-ground bond has almost no visible effect on the DAS records. The plot of the differences, however, shows that the weaker coupling generates a number of different arrivals: surface waves as seen in Section 3.1.1 (although weaker); smaller phases that reflect from the cable's end at channel 250 arriving with opposite travel time slope. Because the only change we introduced is the stiffness of the cable-ground coupling, the observed additional phases are a direct consequence of the weak coupling. We can explain this by the fact the elastic spring parameter K is directly connected to the Lamé parameters of the medium (O'Brien & Bean 2004), so that a change in the elastic bond stiffness is equivalent to a change in the medium's elastic properties (Fig. 10), thus affecting the seismic wavefield.

Cable with its own material properties, weakly coupled to a homogeneous medium: For the simulation where the cable has its own elastic properties (Fig. 8c), the weaker cable-ground coupling has visible effects. The signal is further amplified with respect to the simulation which included the cable's own elastic properties but a perfect cable-ground coupling. Additionally, surface waves are substantially delayed. This effect is particularly visible at more distant DAS channels. Similar to what we observed in Section 3.1.1, the presence of a weakly coupled cable has more visible effects on the surface waves than on body waves.

3.2 Cable trench

After measuring the effects of changes in the cable's properties and its coupling to the medium, we study the effects of the presence of a trench around the cable. This is important as in most real experiments, the cable is either buried in the ground or is deployed in boreholes and is only rarely just lying on the surface or in direct contact with bedrock. After being dug, trenches are refilled with materials that are often unconsolidated, and only rarely are dense, stiff materials such as cement used. This implies that in many applications, the cable is surrounded by, and thus coupled to, a thin, unconsolidated layer. We simulate this in ELM2D-DAS by adding a trench as a one particle (1 m thickness) layer around the cable with its own material and bond properties (Fig. 11). The DAS itself retains the cable properties tested in Section 3.1.1, and in this section is fully coupled to the trench (the combined effects of a cable weakly coupled to a trench will be discussed in Section 3.3).

3.2.1 Slow trench

In the first test we only vary the material properties of the trench, setting $V_P = 1800\ \mathrm{m\,s^{-1}}$, $V_S = 1045\ \mathrm{m\,s^{-1}}$ and $\rho = 1500\ \mathrm{kg\,m^{-3}}$ to simulate unconsolidated sediments (Zimmer *et al.* 2002; Fig. 11 a). The chosen values create a significant velocity contrast with the high V_P, V_S of the homogeneous half-space, that is representative of the strong velocity contrast often found in a variety of DAS

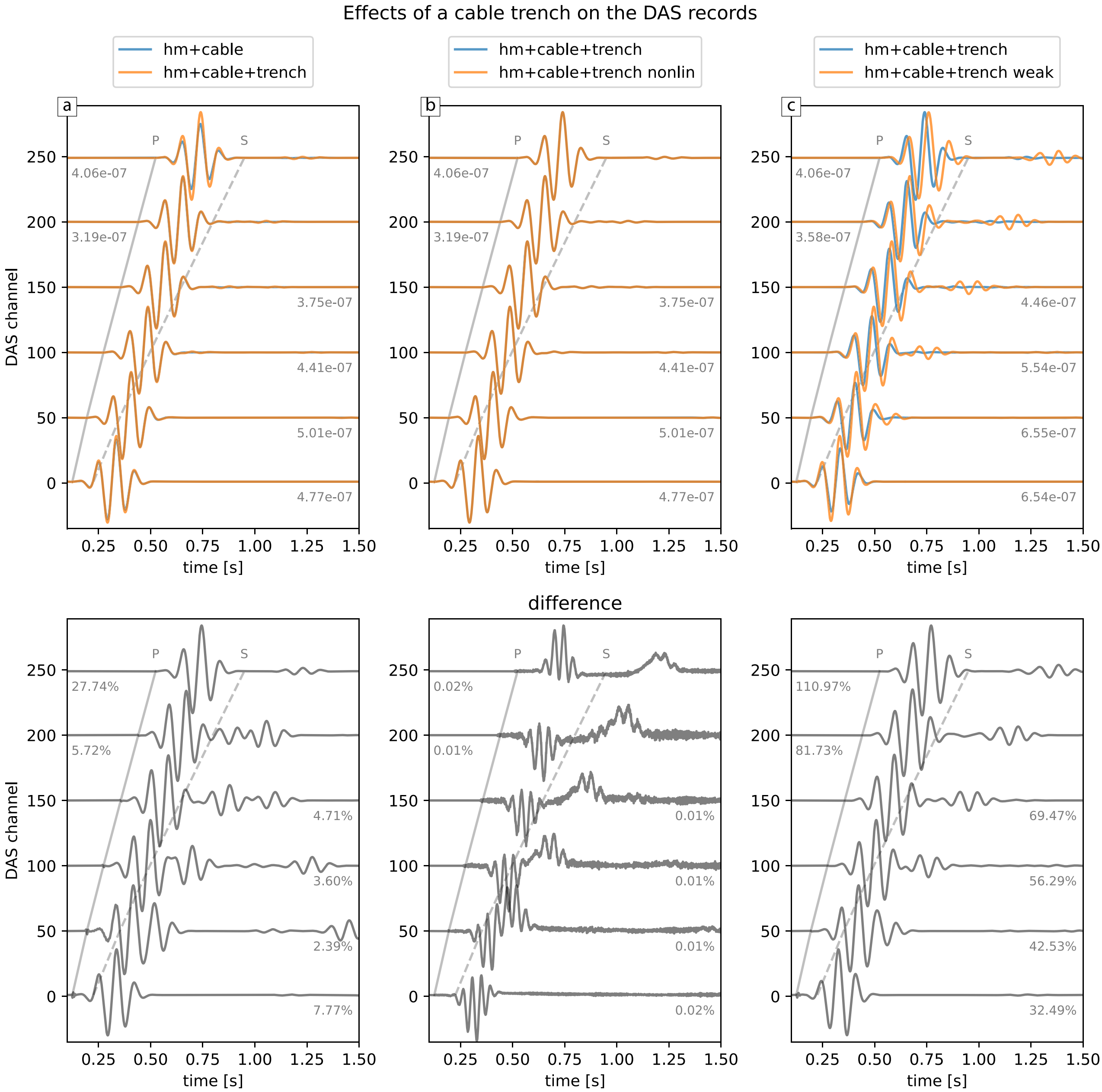

Figure 11. Effect of a cable trench on DAS records. The geometry of the experiment is described in Fig. 7. Each column represents a simulation: top, DAS channel axial strain records (orange), compared to the reference case (blue) (legend for both records plotted on top of each panel); bottom, difference between the two simulations. All channel records are normalized to the maximum axial strain amplitude, which is reported next to each waveform. The amplitudes of the differences are shown as percentage of the axial strain maximum.

deployments (e.g. urban, concrete/bedrock and sediments; borehole, bedrock/drilling hole; glacial, snow/firn and ice/bedrock). Since the trench introduces a low-velocity zone in the model, we would expect a delay of the signal, which we do not observe, likely because of the small width of the trench relative to the seismic wavelength. The last channel of the DAS cable however is visibly amplified compared to the simulation with no trench. Interestingly, this amplification is not visible in further DAS channels. By looking at the plot of the differences, we can clearly see the presence of a signal reflected from the end of the trench. The amplification we observe at the cable's end (channel no. 250) is likely an effect of this reflection interfering constructively with the direct arrival within the last channel gauge.

3.2.2 Non-linear trench

Unconsolidated materials often present non-linear stress-strain relationships. This has been shown to significantly affect the seismic response of the material in both laboratory experiments and numerical simulations (Johnson & Rasolofosaon 1996; Martin *et al.* 2019; O'Brien 2021). We simulate this by adding non-linear springs to the cable trench, using the method described in (O'Brien 2021). We started by using the non-linear coefficients used by (O'Brien 2021) to simulate tens of metres of unconsolidated volcanic scoria layers. These values however generated no change in our simulation, likely because of the small thickness of our non-linear trench. We then

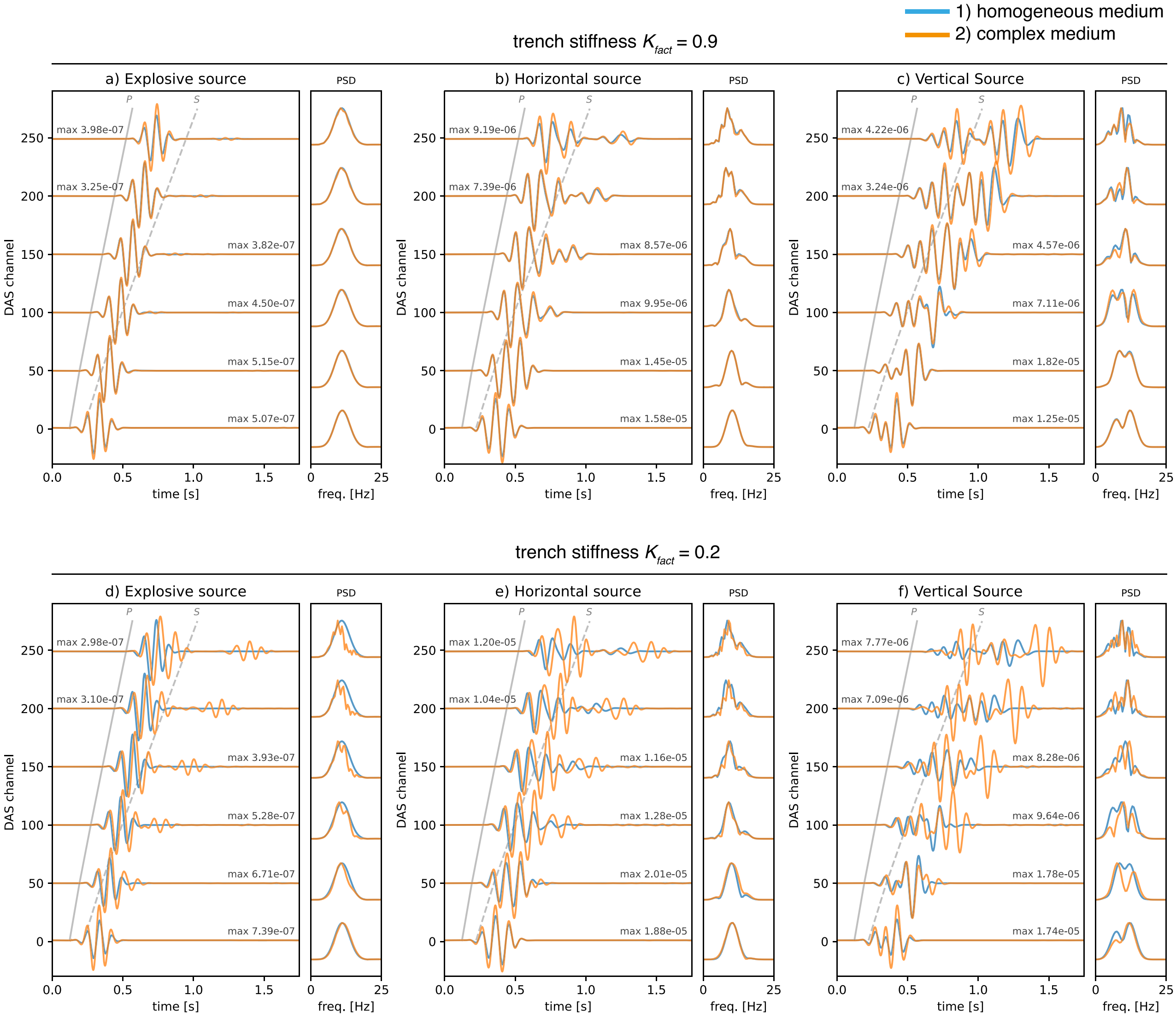

Figure 12. DAS cumulative response effects for different source types. Rows show the results for a complex medium with a mildly (top panel) and strongly (bottom panel) compliant trench. Columns show the results with explosive, horizontal and vertical force sources, respectively. In all panels, traces are coloured following the simulation design: blue, homogeneous half-space; orange, complex medium. Arrival times for the *P* and *S* waves are shown in grey on all panels. All channel records are normalized to the maximum axial strain amplitude, which is reported next to each waveform.

used the same coefficients but scaled up by one order of magnitude ($l_0 = -80e^{11}$ Nm^2, $m_0 = -200e^{11}$ Nm^3) to test if stronger non-linearity could trigger visible effects. These stronger non-linear coefficients produced detectable results that are however just above numerical noise levels (Fig. 11).

The differences between the linear and non-linear trench case shows that non-linearity in the trench adds a very small signal that bears all the characteristics expected in waves propagating in non-linear materials (Martin *et al.* 2019; O'Brien 2021): trace asymmetry and the presence of the first overtone (twice the source frequency). The small amplitudes we observe are likely a consequence of the very thin non-linear layer (1 m) and its distance from the source (>500 m), resulting in negligible effects in our specific setup. However, the characteristic signal we observe as a result of the introduced non-linearity show that non-linear effects—the amplitude of which varies with the signal's amplitude (O'Brien 2021)—can be excited even for very thin, sub-wavelength non-linear heterogeneities.

3.2.3 Trench stiffness

Another common characteristic of unconsolidated materials is that their internal bonds are less stiff than in solid rock. In ELM2D-DAS, we simulate this by reducing the stiffness of the bonds between particles in the trench. This is done in the same way as the cable-ground coupling in Section 3.1.2, but instead of just reducing the stiffness of the interface between different media (making it more compliant), we reduce the stiffness of the bonds between all particles forming the trench.

The elastic properties of sediments can vary greatly with their strength: the Young's modulus of loose sands and gravels can be as low as 10 per cent of its value for densely packed samples, and even lower when comparing soft and stiff clays and silts (Obrzud & Truty 2012). We can translate such changes in elastic parameters into the spring stiffness of our lattice using the Lamé constants (O'Brien & Bean 2004). To simulate a compliant trench, we decrease the stiffness of the bonds between the trench particles by 80%

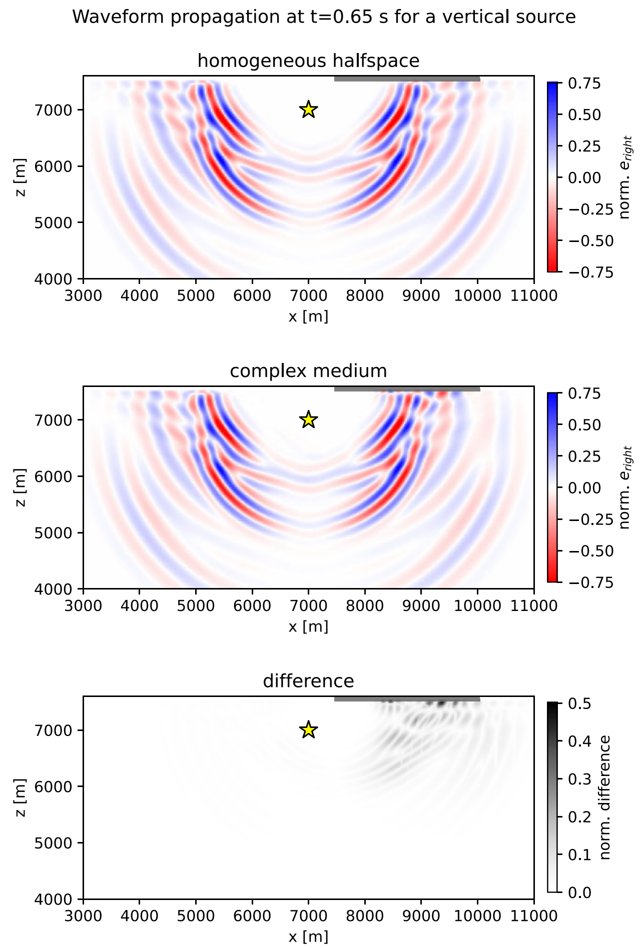

Figure 13. Effects of the complex instrument response scenarios on the propagating strain wavefield. Top panels show the right-bond strain at $t = 0.65$ s for the simulation described in Fig. 12(f): homogeneous medium and complex model including all cable-ground complexities. Bottom panel shows the absolute difference between the two. All colourmaps are normalized to the maximum. In all panels, the source location is shown as a yellow star and the DAS as a dark grey line.

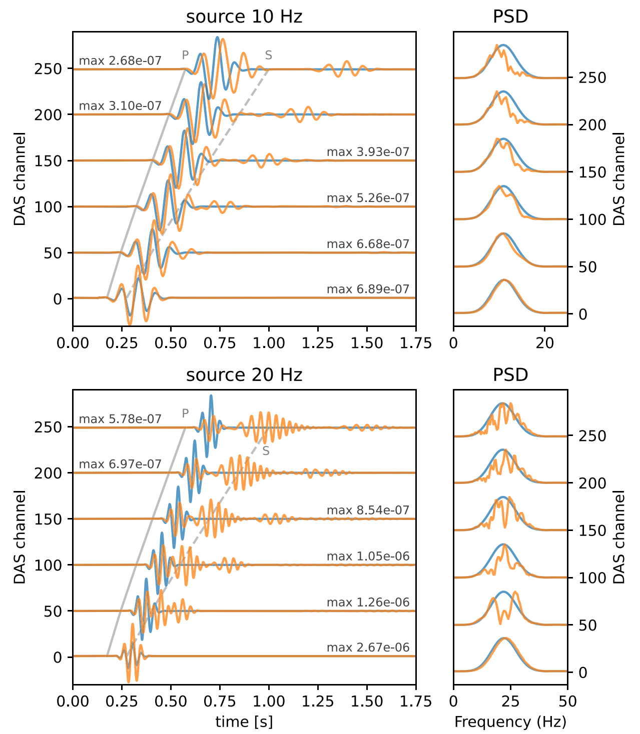

Figure 14. DAS cumulative response effects for different frequency sources. Top panels show the results for a 10 Hz source, bottom panels for a 20 Hz source. In all panels, traces are coloured following the simulation design: blue, homogeneous half-space; orange, complex medium. Arrival times for the *P* and *S* waves are shown in grey on all panels. All channel records are normalized to the maximum axial strain amplitude, which is reported next to each waveform.

($K_{\mathrm{fact}} = 0.2$), which implies a 71% decrease in the Young's modulus (Fig. 10).

In Fig. 11, we observe that by strongly reducing the stiffness of the bonds between the trench particles, the DAS records get amplified and significantly delayed. Additionally, the surface waves are much more amplified and delayed compared to the body wave arrivals. As discussed previously for the case of a weak cable-ground coupling, the spring constant K is directly linked to the elastic parameters of the medium (O'Brien & Bean 2004), so that decreasing the spring stiffness results in decreasing seismic wave velocity (Fig. 10). Less stiff trench materials thus effectively translate into an even lower-velocity trench in our simulation, which explains the amplifications and delays we observe. Our results indicate that the state of the material to which the cable is directly coupled is one of the key factors affecting the response of a DAS cable.

3.3 Cumulative cable response effects

In the previous sections, we have analysed the effects of each change independently. We observe that many of the tested ground-coupling scenarios have subtle effect on the DAS records, with the exception of the stiffness of the trench material, which has by far the largest effect.

If we compare the simulations in a homogeneous half-space to one in which we include all the previously individually tested scenarios—cable has its own material properties and is poorly coupled to a trench that is also non-linear and compliant (referred as 'complex medium' from here onwards)—we can see that the cumulative simulated variations in the instrument response have clear, visible effects on the DAS record.

Observing the simulations for an explosive source (Figs 12a and d), we can again see that the stiffness of the trench material has the strongest effect, introducing amplification, the excitation of surface waves and strong delays in the DAS records. Notably, the aforementioned effects get substantially more pronounced if we simulate a vertically or horizontally polarized source (Figs 12b, c, e and f). For polarized sources, the recorded wave train—already more complex due to the additional presence of shear waves—is strongly affected by the addition of the cumulative ground-coupling complexities, showing substantial amplification of the head- and surface wave phases. By observing snapshots of the strain field for a vertical source (Fig. 13), we can see that the largest differences in the wavefield from the homogeneous half-space simulation are indeed located next to the surface, while the wavefield at depth is minorly affected. For the case of the vertical source in particular, the DAS traces recorded in the medium including all the cumulative ground-coupling effects bear very little similarity to the ones recorded in the homogeneous medium with an identical geometry (Fig. 12f).

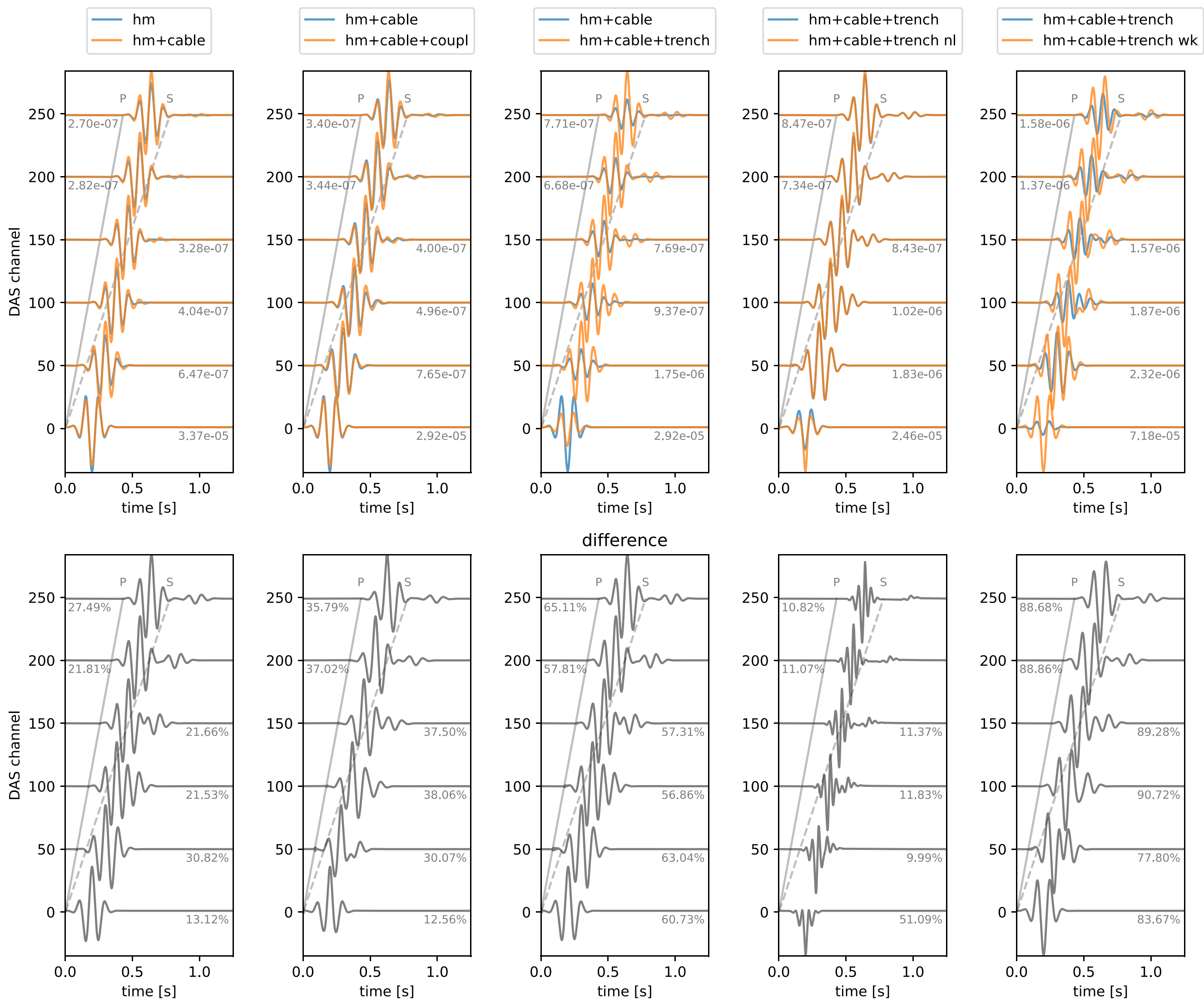

Figure 15. Effects of changes in the cable coupling and local site effects for a source on the cable's end. Each column represents a simulation: top, DAS channel axial strain records (orange), compared to the reference case (blue) (legend for both records plotted on top of each panel); bottom, difference between the two simulations. All channel records are normalized to the maximum axial strain amplitude, which is reported next to each waveform. The amplitudes of the differences are shown as percentage of the axial strain maximum.

3.3.1 Higher source frequency effects

In the previous sections, we used the same 10 Hz source across all simulations for consistency. To further test the effects of the modelled changes in the cable coupling and trench properties, we compare the same simulation shown in Fig. 12(d) (explosive source 500 m to the left and below the cable's start, homogeneous half-space versus complex medium) to the same setup with a 20 Hz Morlet wavelet source (Fig. 14). It is important to note that for both frequencies the trench and the cable have sub-wavelength thickness.

In the 20 Hz simulation, we observe the same delay of the *P* wave arrival as seen for the 10 Hz Morlet source. We also see the surface waves arriving late in the trace. What is immediately apparent however, is the presence of a strong, almost monochromatic, and slightly dispersive signal in between the *P* and Rayleigh wave arrivals for the 20 Hz simulation. Due to these characteristics, we can identify this as an interface wave propagating along the discontinuity formed by the slow, compliant, and thin trench layer as it would do along a fracture (Toomey 2001). Interestingly, this phase is not visible in the 10 Hz simulation, although both sources have a wavelength much larger than the trench thickness, even when considering the slowest S wave speeds (1045 m s^{-1}) and the slowing effect of the decreased coupling and trench stiffness (Fig. 10).

3.4 Source on the cable

Until now, we saw that most scenarios we tested (e.g. cable material properties, cable-ground coupling, slow trench, trench nonlinearity) have relatively small effects on the DAS records. This changes dramatically if we put the source on one of the cable's ends: in Fig. 15, we tested the same set of scenarios as in Figs 8 and 11, but with a cable in the middle of a homogeneous medium, no free surface and an explosive source at the cable's start. What we observe is that in this scenario, all changes contribute visibly to different degrees of amplification of the signal, which can be explained by

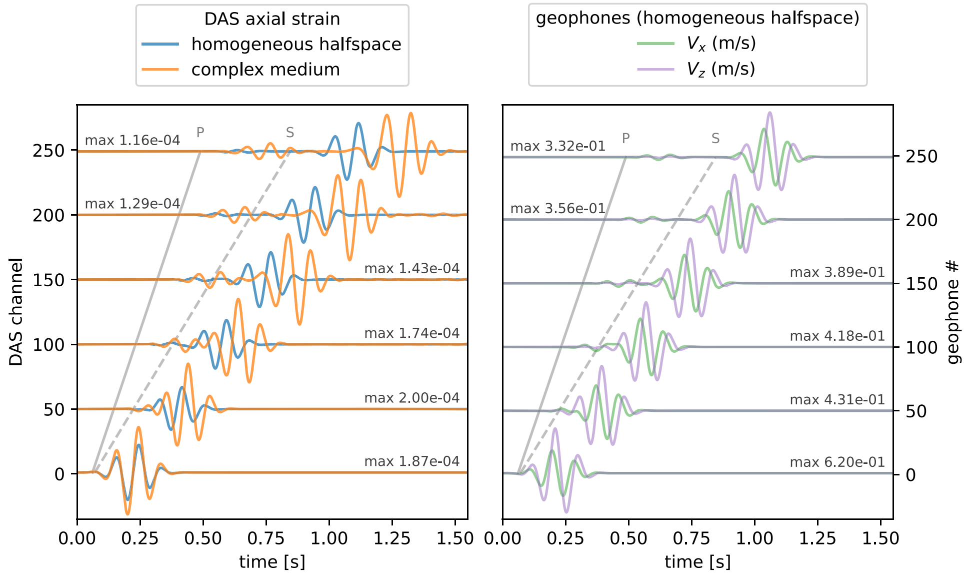

Figure 16. Comparison between DAS and geophones for a vertical source at the surface. DAS records show axial strain and are coloured following the simulation design: blue, homogeneous half-space; orange, complex medium including the cable and trench with the complex properties discussed in the text. Geophone records are computed in the homogeneous half-space only as they do not require a trench, and show velocity, coloured following the component direction: green, horizontal; purple, vertical. Arrival times for the *P* and *S* waves are shown in grey on all panels. All channel records are normalized to the maximum axial strain or velocity amplitude, which is reported next to each waveform.

the fact that the source itself is now located on the cable, so that the strong near-field energy is directly injected into—and perhaps guided along—the weakly coupled cable or compliant trench.

3.5 Source at the surface and comparison to geophones

In previous sections we have seen that the scenarios we modelled have stronger effects on surface waves compared to body waves. Due to the source location at depth, however, the simulations discussed so far do not generate strong surface waves. In Fig. 16, we retain the same properties shown in Fig. 12(f) ('complex medium': cable with its own material properties, weak cable-ground coupling, and a slow, compliant, non-linear trench), but use a vertical source located at the surface. This setup is of particular interest as it is representative of the geometry used in many active source seismic experiments. The resulting virtual DAS records show how the surface wave train is substantially amplified, delayed and distorted. Compared to what we observed in Fig. 12(f) for a source at depth, the overall waveform is simpler due to the wave front travelling parallel to the DAS cable's axis, showing more clearly the changes in the surface wave arrival, such as the longer and more complex wave shape. This simulation shows that the cumulative effects of the simulated cable material and ground coupling, and the properties of the surrounding trench cause complex changes to the recorded waveforms that go beyond simple amplification and delay.

In Fig. 16 we also compare the axial strain recorded by the virtual DAS to the horizontal and vertical seismic velocity recorded by virtual geophones collocated with the cable's channels at the surface. Since geophones do not require to be deployed in a trench, we model them as perfectly coupled to the surface of the homogeneous. The response of most geophones is flat at 10 Hz—the frequency of our signal—so we do not apply a response function to the virtual geophone records. As expected, the geophone records in Fig. 16 closely resemble the DAS axial strain in the homogeneous half-space, with the different wave shape explained by the different measured quantity (x, z velocity versus axial strain). This comparison clearly shows us that the deployment of the DAS—and the trench containing it—significantly alters the wavefield that would be recorded in a traditional deployment using seismometers.

4 CONCLUSIONS

In this study, we used numerical simulations to model the response of a DAS cable and quantitatively analyse the effects of instrument coupling and local site effects on the strain recorded in separate simulations. In particular, we explored the effects of the cable-ground coupling and the properties of the optical fibre cable and the material immediately surrounding it.

Our simulations show that changing these properties can strongly change the amplitude and arrival time of the signal we simulate. Our results clearly highlight that the properties of the material in the immediate surroundings of the cable—and especially its stiffness—have the largest effect on the DAS records, resulting in: amplifications as large as twice the wavelet amplitude; large phase delays, especially for surface waves; selective amplification of surface wave phases; generation of interface waves for higher source frequencies. If we consider that these effects can, in practice, vary along the cable, our study shows that cable ground coupling and local site effects are important, and that particular attention should be paid to the properties of the material to which the optical fibre cable is coupled.

Our results also show changes in the frequency of the signal only for the case of a non-linear trench. In all other tested scenarios, the spectra of the DAS records are much more affected by other factors such as the interference of different parts of the wave train. This is in agreement with what was observed empirically in previous studies (Paitz *et al.* 2021).

The simulations presented in this study represent a simplification of a real DAS experiment. Our observations however, confirm quantitatively that the amplitude variations observed in empirical studies on DAS response—until now hypothetically attributed to either coupling variations along the fibre (Harmon *et al.* 2022) or small, local heterogeneities (Paitz *et al.* 2021)—are likely to be a consequence of less stiff bonding either between the cable and the medium surrounding it, or, even more likely, within the medium to which the cable is coupled. We suggest that the future development and application of tools such as ELM2D-DAS capable of simulating the details of dynamic strain propagation in the vicinity of the receiver could greatly improve our understanding of strain records from optical fibre cables.

DATA/CODE AVAILABILITY

We make ELM2D-DAS available for download to the scientific community on git (https://git.dias.ie/seismology/elm2d-das). ELM2D-DAS is a flexible numerical forward modelling tool that can simulate the full strain- and displacement fields in 2-D and the corresponding synthetic DAS and seismometer records. Its granular control on material properties can be used to simulate instrument coupling, local site effects as well as other complex material properties such as non-linearity and fractures. It consists of a set of C programs running the simulation, with a python wrapper that allows quick, easy changes in the experiment setup. The code uses MPI for parallelization and has been successfully tested on Linux with up to 150 cores and in distributed computing with two parallel servers.

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AUTHOR CONTRIBUTIONS

NC and CB jointly conceived the paper. NC developed the new version of the code, designed and ran the simulations and made the figures. GOB provided the original ELM code. NC wrote the manuscript with the help of CB. All authors contributed to discussion.

CONFLICT OF INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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